

AI and Machine Learning - Study Notes

1. What is Artificial Intelligence?

Artificial Intelligence (AI) refers to the simulation of human intelligence processes by machines, especially computer systems. These processes include learning, reasoning, and self-correction. AI was formally founded as a discipline in 1956 at a Dartmouth College workshop.

- * AI enables machines to mimic human cognitive functions such as learning and problem-solving.
- * It is used in robotics, natural language processing, computer vision, and more.
- * Types of AI: Narrow AI (ANI), General AI (AGI), and Super AI (ASI).

2. Machine Learning - Core Concepts

Machine Learning (ML) is a subset of AI that provides systems the ability to automatically learn and improve from experience without being explicitly programmed. Arthur Samuel coined the term Machine Learning in 1959. ML focuses on developing programs that learn from data.

- * Supervised Learning: The model is trained on labeled data for classification or regression.
- * Unsupervised Learning: The model finds patterns in unlabeled data such as clustering.
- * Reinforcement Learning: An agent learns by interacting with an environment and receiving rewards.
- * Semi-supervised Learning: Uses a small amount of labeled and a large amount of unlabeled data.

3. Neural Networks and Deep Learning

Deep Learning is a subfield of machine learning that uses neural networks with many layers to learn representations of data with multiple levels of abstraction. Geoffrey Hinton, Yann LeCun, and Yoshua Bengio shared the 2018 Turing Award for their contributions to deep learning.

- * Perceptron: The simplest neural network unit. Takes inputs, applies weights, outputs a signal.
- * Activation functions: ReLU, Sigmoid, Tanh introduce non-linearity into the network.
- * Backpropagation: Algorithm to update weights by computing the gradient of the loss function.
- * CNN: Convolutional Neural Networks that excel at image recognition tasks.
- * RNN and LSTM: Recurrent networks that excel at sequential data like text and time series.

4. Natural Language Processing (NLP)

Natural Language Processing (NLP) is the branch of AI concerned with giving computers the ability to understand, interpret, and generate human language. NLP combines computational linguistics with machine learning and deep learning. Key tasks include tokenization, named entity recognition, sentiment analysis, machine translation, and question answering.

- * Tokenization: Breaking text into smaller units called tokens (words or subwords).
- * Embeddings: Dense numerical representations of words in a continuous vector space.
- * Word2Vec: A model that learns word associations from large text corpora.
- * BERT: Bidirectional Encoder Representations from Transformers, Google's 2018 breakthrough.
- * GPT: Generative Pre-trained Transformer by OpenAI, used for text generation.

5. The Transformer Architecture

The Transformer architecture was introduced in the paper Attention Is All You Need by Vaswani et al. in 2017. It revolutionized NLP and AI by using self-attention mechanisms to process sequences in parallel, making training much faster than RNNs. It has two main parts: Encoder and Decoder.

- * Self-Attention: Allows each token to attend to every other token in the sequence.
- * Multi-Head Attention: Runs multiple attention heads in parallel for richer representations.
- * Positional Encoding: Injects sequence order information since Transformers have no recurrence.
- * Feed-Forward Layer: Applied after attention to further transform the representations.

6. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is a technique that enhances Large Language Models (LLMs) by retrieving relevant information from an external knowledge base before generating an answer. RAG reduces hallucinations and was introduced by Lewis et al. from Facebook AI Research in 2020.

- * Indexing: Documents are chunked, embedded, and stored in a vector database.
- * Retrieval: The user query is embedded and the top-K most similar chunks are fetched.
- * Generation: The LLM generates an answer conditioned on the retrieved chunks.
- * FAISS: Facebook AI Similarity Search, an efficient library for dense vector retrieval.
- * Benefits: Grounded answers, reduced hallucination, no retraining needed for new data.

7. Model Evaluation Metrics

Evaluating machine learning models requires choosing the right metrics depending on the task. Using the wrong metric can give a misleading picture of performance, especially on imbalanced datasets.

- * Accuracy: Percentage of correct predictions. Good for balanced classes.
- * Precision: Of all positive predictions, what fraction were actually positive?
- * Recall (Sensitivity): Of all actual positives, what fraction were correctly identified?
- * F1-Score: Harmonic mean of precision and recall. Good for imbalanced datasets.
- * ROC-AUC: Area under the ROC curve. Measures the discrimination ability of a classifier.
- * BLEU Score: Used to evaluate machine translation and text generation quality.
- * Perplexity: Measures how well a language model predicts a sample. Lower is better.

8. AI Ethics and Responsible AI

As AI systems become more powerful and widely deployed, ethical considerations have become critically important. Responsible AI development requires addressing fairness, accountability, transparency, and privacy. The EU AI Act (2024) is the world's first comprehensive legal framework for AI systems.

- * Bias and Fairness: AI systems can inherit biases present in training data.
- * Explainability (XAI): The ability to explain why a model made a particular prediction.
- * Privacy: Models trained on personal data risk exposing sensitive information.
- * Hallucination: LLMs sometimes generate confident but factually incorrect information.
- * Alignment: Ensuring AI systems pursue goals that are aligned with human values.